RAJALAKSHMI ENGINEERING COLLEGE

Sagana R 251001223-2025-IT

S2

Started on	Friday, 21 August 2026, 2:03 PM
State	Finished
Completed on	Friday, 21 August 2026, 2:15 PM
Time taken	11 mins 28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    int sum=0;
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    scanf("%d",&sum);
    int l=0,r=n-1,f=0;
    while(l<r){
        int c=a[l]+a[r];
        if(c==sum){
            printf("%d\n",a[l]);
            printf("%d\n",a[r]);
            f=1;
            break;
        }
        else if(c>sum){
            r--;
        }
        else {
            l++;
        }
    }
    if(!f){
        printf("No\n");
    }
    return 0;

}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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